b230e5.c

```c
#include <stdio.h>
#include <stdlib.h>

typedef struct node
{
    int data;
    struct node *next;
} node;

node *Insert_beg(node *head, int x)
{
    node *p = (node *)malloc(sizeof(node));
    p->data = x;
    p->next = head;
    head = p;
    printf("Inserted Sensor Reading (%d%%) at Beginning.\n", x);
    return head;
}

node *Insert_end(node *head, int x)
{
    node *p = (node *)malloc(sizeof(node));
    p->data = x;
    p->next = NULL;

    if (head == NULL)
    {
        return p;
    }

    node *temp = head;
    while (temp->next != NULL)
    {
        temp = temp->next;
    }
    temp->next = p;
    printf("Appended Sensor Reading (%d%%) at End.\n", x);
```

b230e5.c

b230e5.c > main()

```c
node *Insert_end(node *head, int x)
    return head;
}

node *Delete_beg(node *head)
{
    if (head == NULL)
    {
        printf("Empty Linked List. Nothing to delete.\n");
        return head;
    }
    node *p = head;
    printf("Deleted Sensor Reading from Beginning: %d%%\n", p->data);
    head = head->next;
    free(p);
    return head;
}

void PrintList(node *head)
{
    if (head == NULL)
    {
        printf("List Status: [ Empty ]\n\n");
        return;
    }

    node *p = head;
    printf("Current Soil Moisture Sensor Stream: ");
    while (p != NULL)
    {
        printf("%d%% -> ", p->data);
        p = p->next;
    }
    printf("NULL\n\n");
}

int main()
```

b230e5.c

```c
int main()
{
    node *head = NULL;

    printf("=== SUGARCANE IRRIGATION SYSTEM (SINGLY LINKED LIST) ===\n\n");

    printf("1. Logging regular soil moisture sensor readings at End...\n");
    head = Insert_end(head, 42);
    PrintList(head);

    head = Insert_end(head, 35);
    PrintList(head);

    head = Insert_end(head, 28);
    PrintList(head);

    printf("2. Logging critical soil moisture reading at Beginning (High Priority)...\n");
    head = Insert_beg(head, 12);
    PrintList(head);

    printf("3. Processing highest priority sensor node from Beginning...\n");
    head = Delete_beg(head);
    PrintList(head);

    return 0;
}
```

```
=== SUGARCANE IRRIGATION SYSTEM (SINGLY LINKED LIST) ===

1. Logging regular soil moisture sensor readings at End...
Current Soil Moisture Sensor Stream: 42% -> NULL

Appended Sensor Reading (35%) at End.
Current Soil Moisture Sensor Stream: 42% -> 35% -> NULL

Appended Sensor Reading (28%) at End.
Current Soil Moisture Sensor Stream: 42% -> 35% -> 28% -> NULL

2. Logging critical soil moisture reading at Beginning (High Priority)...
Inserted Sensor Reading (12%) at Beginning.
Current Soil Moisture Sensor Stream: 12% -> 42% -> 35% -> 28% -> NULL

3. Processing highest priority sensor node from Beginning...
Deleted Sensor Reading from Beginning: 12%
Current Soil Moisture Sensor Stream: 42% -> 35% -> 28% -> NULL
```